Lecture 03 – 04

Chapter 3: Counting Processes

References:

1. Oliver Ibe – Markov Processes for Stochastic Modeling (Chapter 2)
2. Sheldon Ross – Introduction to Probability Models (Chapter 5)

Objectives:

1. Probability Model of two very simple Counting processes
2. Arrival pattern of packets in a router
3. Traffic generation in communication node

Counting Processes:

1. Poisson Process
2. Bernoulli Process

Stochastic Process

1. An SP is a collection of random variables
    a. May be infinite in number
2. Each RV is a time sample of the SP at time t, when t is fixed
3. The parameter time can be discrete or continuous
    a. An SP is said to be discrete- or continuous-time SP
4. If both t and w are fixed, the value of the SP is a real number
    a. Represents the state of the SP at t
5. Set of all possible states of the SP is
    a. Either discrete or continuous, known as the state-space of the SP

We usually denote a SP by either

- $\{X(t), t \geq 0\}$, if it is a continuous-time process
- $\{N_i, i = 0, 1, \dots\}$, if it is discrete -time process

Classification of the Stochastic Processes

1. Discrete-time and discrete-state SP
2. Discrete-time and continuous-state SP
3. Continuous-time and discrete-state SP
4. Continuous-time and continuous-state SP

Counting Process (CP)

An SP $\{X(t), t \geq 0\}$ or $\{N_i, i = 0, 1, \dots\}$ is called a counting process (CP), if $X(t)$ or N_i represents

- The number of arrivals of a given event, and
- This number of arrivals is counted up to time t (in the interval $[0, t]$)

Examples:

1. If $X(t)$ is the number of persons who enter a particular store at or prior to time t, $\{X(t), t \geq 0\}$ is a CP

2. If an event occurs whenever a child is born, then $\{X(t), t \geq 0\}$ is a counting process when $X(t)$ equals the total number of babies born by time t.
   a. Does $X(t)$ include persons who have died by time t? Explain why it must.

3. If X(t) equals the number of goals a soccer player scores by time t, then $\{X(t), t \geq 0\}$ is a counting process. An event of this process will occur whenever the soccer player scores a goal.

CP properties

- Starts counting from 0
  $X(t) = 0$, if $t < 0$

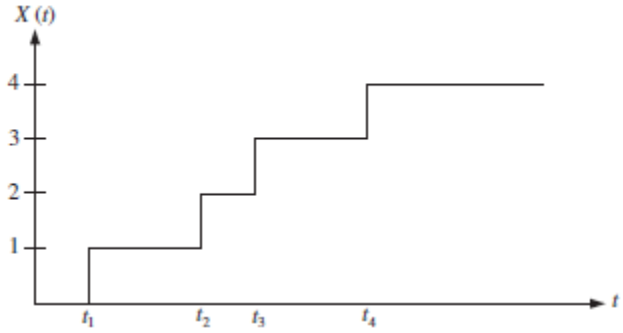
- $X(t)$ is non-negative
  $X(t) \geq 0$

- $X(t)$ is integer valued

- $X(t)$ is a non-decreasing function of time
  If $s < t, X(s) \leq X(t)$

- Number of arrivals in $(s, t]$ is given by
  $X_{s,t} = X(t) - X(s)$

Sample path of a CP



Stationary Stochastic Process

- An SP is stationary, if its statistical properties do not change with time
  - The SP is time invariant
- Joint probability distributions (or multivariate distributions) do not change by a time shift
- Mean or variance stays the same
- Correlation depends on time difference, not on actual time

Strict-Sense Stationary (SSS) SP

An SP is SSS if

$$f_X(x_1, \dots, x_n; t_1, \dots t_n) = f_X(x_1, \dots, x_n; t_1 + \tau, \dots t_n + \tau), \text{ or}$$

$$P_X(x_1, \dots, x_n; t_1, \dots t_n) = P_X(x_1, \dots, x_n; t_1 + \tau, \dots t_n + \tau)$$

For every

- Integer n
- Time instants $t_1, \dots, t_n$
- Time shift τ

Wide -sense Stationary (WSS) SP

An SP is WSS, if

- Mean is constant
  $E[X(t) = \mu$, independent of t
- Autocorrelation depends only on time difference
  $R_X(t_1, t_2) = R_X(t_1, t_1 + \tau) = R_X(\tau)$

Examples:

WSS: $X(t) = A\sin(\omega t + \Theta)$

Mean is zero and autocorrelation depends on time shift

SSS $X(t) \sim N(0, \sigma^2)$

Finite dimensional and joint distributions are identical

Characteristics of CPs

<u>Independent Increment Process (IIP)</u>

- The number of arrivals in two disjoint time intervals are independent
- A CP possesses the independent increment property, if
  - The number of arrivals in disjoint periods are independent
  - For every set of time intervals $T = \{\mathrm{t}_1, \mathrm{t}_2, \dots\}$, where $0 < t_1 < t_2, \dots, < t_n$, the RVs

$$X_1 = X(t_1) - X(t_0)$$
$$X_2 = X(t_2) - X(t_1)$$

$$X_n = X(t_n) - X(t_{n-1})$$

are independent, the CP has independent increments

Stationary Increment Process (SIP)

The number of arrivals in any time interval depends only on the length of the interval not on the actual time

The distribution of the number of arrivals is same for same intervals

A CP $\{X(t), t \geq 0\}$ is a SIP, if for every set of equal time intervals

- The number of arrivals in each interval are identically distributed RVs
- For time intervals of equal lengths
  $(t_1, t_1 + s), (t_2, t_2 + s), \dots, (t_n, t_n + s)$
- Let the RVs are
  $X_1 = X(t_1 + s) - X(t_1)$
  $X_2 = X(t_2 + s) - X(2)$

  $X_n = X(t_n + s) - X(t_n)$
- If $X_1, X_2, \dots, X_n$ are identically distributed, then $X(t)$ is a SIP

<u>An alternate way of defining an SIP is</u>

For a CP $\{X(t), t \geq 0\}$,

If the distribution of the RV representing the number of arrivals in $(t, t+s)$

- does not depend on t, but
- only depends on s

Then $X(t)$ is an SIP.

The RV is

$X = X(t+s) - X(t)$

If the distribution of X does not depend on t, $X(t)$ is a SIP.

Both definitions are SSS.

A CP $X(t)$ is a SIP, in the sense of WSS, if the 1st or 2nd order statistics are independent of t

The mean of a CP $X(t)$ is SIP, in the sense of WSS, if

$$E[X(t)] = m$$

The constant m is the value of the mean at time $t = 1$,

$$m = E[X(1)]$$

And the variance

$$Var[X(t)] = \sigma^2$$

The constant σ^2 is the value of the variance at time $t = 1$

$$\sigma^2 = V[X(1)]$$

Poisson Process

<u>Definition</u>

A CP $\{X(t), t \geq 0\}$, where the number of arrivals in any interval of length t is Poisson distributed, with

$\lambda \rightarrow$ average arrival rate
$\alpha \rightarrow$ average arrival in $t, \alpha = \lambda t$

is a Poisson process

For all $t, s \geq 0$,

$$P[X(s+t) - X(s) = n] = \frac{(\lambda t)^n}{n!} e^{-\lambda t} \quad n = 0, 1, 2, \ldots$$

If <u>λ is independent of time</u>, the Poisson process is <u>homogeneous</u>, otherwise it is non-homogeneous and the arrival rate is $\lambda(t)$

<u>Alternate Definition</u>

If $\{X(t), t \geq 0\}$ is a CP with stationary and independent increment and rate $\lambda > 0$

1. Probability of one arrival in a small-time interval Δt is

$$P[X(t+\Delta t) - X(t) = 1] = \lambda \Delta t + o(\Delta t)$$

$$\lim_{\Delta t \to 0} \frac{o(\Delta t)}{\Delta t} = 0$$

2. Probability of 2 or more arrivals in Δt

$$P[X(t+\Delta t) - X(t) \geq 2] = o(\Delta t)$$

3. Prob. of zero arrival in Δt

$$P[X(t+\Delta t) - X(t) \geq 0] = 1 - \lambda \Delta t + o(\Delta t)$$

For any Δt, $N(\Delta t) \sim \text{Pois}\,(\lambda \Delta t)$. Hence

$$P\{N(\Delta t) = 0\} = e^{-\lambda \Delta t}, P\{N(\Delta t) = 1\} = e^{-\lambda \Delta t}(\lambda \Delta t),$$

$$P\{N(\Delta t) \geq 2\} = 1 - P\{N(\Delta t) = 0\} - P\{N(\Delta t) = 1\} = 1 - e^{-\lambda \Delta t}(1 + \lambda \Delta t).$$

Plugging into the formulas:

- $P\{N(\Delta t) = 0\} = 1 - \lambda \Delta t + O((\Delta t)^2).$
  $= 1 - \lambda \Delta t + o(\Delta t).$
- $P\{N(\Delta t) = 1\} = (\lambda \Delta t)(1 - \lambda \Delta t + O((\Delta t)^2)) = \lambda \Delta t + O((\Delta t)^2).$
  $= \lambda \Delta t + o(\Delta t).$

$P\{N(\Delta t) \geq 2\} = 1 - (1 - \lambda \Delta t + O((\Delta t)^2)) - (\lambda \Delta t + O((\Delta t)^2)) = O((\Delta t)^2).$
$= o(\Delta t).$

Then the PMF of the number of events in a time interval of length t

$$
\begin{aligned}
P[X(t+\Delta t)=n] &= P[X(t)=n]P[X(\Delta t)=0] \\
&\quad + P[X(t)=n-1]P[X(\Delta t)=1] \\
&= P[X(t)=n](1-\lambda\Delta t) \\
&\quad + P[X(t)=n-1]\lambda\,\Delta t
\end{aligned}
$$

$$
p_{X(t)}(n,t) = \frac{(\lambda t)^n}{n!}\mathrm{e}^{-\lambda t} \qquad t \geq 0, \qquad n = 0, 1, 2, \ldots
$$

$$
\begin{aligned}
P[X(t+\Delta t)=n] &= P[X(t)=n]P[X(\Delta t)=0] \\
&\quad + P[X(t)=n-1]P[X(\Delta t)=1] \\
&= P[X(t)=n](1-\lambda\Delta t) \\
&\quad + P[X(t)=n-1]\lambda\,\Delta t \\
P[X(t+\Delta t)=n]-P[X(t)=n] &= -\lambda P[X(t)=n]\Delta t \\
&\quad + \lambda\,P[X(t)=n-1]\Delta t \\
\frac{P[X(t+\Delta t)=n]-P[X(t)=n]}{\Delta t} &= -\lambda\,P[X(t)=n]+\lambda\,P[X(t)=n-1]
\end{aligned}
$$

$$
\begin{aligned}
\lim_{\Delta t\to 0}\left\{\frac{P[X(t+\Delta t)=n]-P[X(t)=n]}{\Delta t}\right\} &= \frac{\mathrm{d}}{\mathrm{d}t}P[X(t)=n]=-\lambda\,P[X(t)=n] \\
&\quad + \lambda\,P[X(t)=n-1] \\
\frac{\mathrm{d}}{\mathrm{d}t}P[X(t)=n]+\lambda\,P[X(t)=n] &= \lambda\,P[X(t)=n-1]
\end{aligned}
$$

The last equation can be solved iteratively for $n = 0, 1, 2, \ldots$, subject to the initial conditions

$$
P[X(0)=n]=\begin{cases} 1 & n=0 \\ 0 & n\neq 0 \end{cases}
$$

This gives the PMF of the number of events (or "arrivals") in an interval of length t as

$$
p_{X(t)}(n,t)=\frac{(\lambda t)^n}{n!}\mathrm{e}^{-\lambda t} \quad t\geq 0, \quad n=0,1,2,\ldots
$$

Interarrival time of the Poisson Process

Let, the time of first event is T_1

- For $n > 1$, let T_n denote the time between $(n-1)$-th and the n-th arrivals
- $\{T_n, n = 1,2, \dots\}$ is sequence of interarrival times
- Event $\{T_1 > t\}$ takes place iif no arrival occurs in $[0, t]$

$$P[T_1 > t] = P[X(t) = 0] = e^{-\lambda t}$$

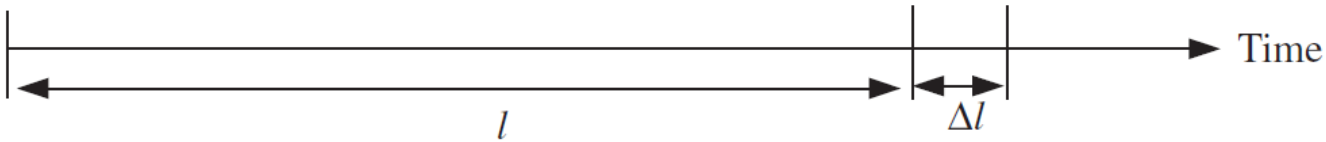
- T_1 has an exponential distribution with rate λ

Similarly, the distribution of T_2 is

$$P[T_2 > t \,|T_1 > s] = P[0 \text{ event in } (s, s+t]|T_1 = s]$$
$$= P\big[0 \text{ event in } (s, s+t]\big] = e^{-\lambda t}$$

$T_n, n = 1, 2, \ldots$ are iid exponential RVs with mean $1/\lambda$

The r-th Arrival Time



$$
\begin{aligned}
f_{L_r}(l)\Delta l &= P[l < L_r \le l + \Delta l] = P[\{X(l) = r-1\} \cap \{X(\Delta l) = 1\}] \\
&= P[X(l) = r-1]P[X(\Delta l) = 1] \\
&= \left\{ \frac{(\lambda l)^{r-1}}{(r-1)!} \mathrm{e}^{-\lambda l} \right\} \{\lambda\ \Delta l\} = \frac{\lambda^r l^{r-1}}{(r-1)!} \mathrm{e}^{-\lambda l}\ \Delta l
\end{aligned}
$$

$$
f_{L_r}(l) = \frac{\lambda^r l^{r-1}}{(r-1)!} \mathrm{e}^{-\lambda l} \quad l \ge 0; \quad r = 1, 2, \ldots
$$

Erlang-r distribution

Compound Poisson Process

$\{N(t), t \geq 0\}$ be a Poisson process with rate λ

$\{Y_i, i = 1, 2, \dots\}$ family of iid RV

Both are independent

Then, $\{X(t), t \geq 0\}$ is a compound Poisson process if

$$X(t) = \sum_{i=1}^{N(t)} Y_i$$

Example:

- Assume students arrive at the university bookstore to buy books in a Poisson manner
- If the number of books that each of these students buys is an IID random variable, then the number of books bought by time t is a compound Poisson process.

The mean and variance of $X(t)$ can be found using the Wald's Equation

$$E\left(\sum_{i=1}^{N} X_i\right) = E(N)E(X)$$

$$\mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) = E(N)\mathrm{Var}(X) + [E(X)]^2\,\mathrm{Var}(N)$$

.

The mean and variance for the compound Poisson Process

$$E[X(t) = \lambda t\, E[Y]$$

$$\sigma^2 = \lambda t * \sigma_Y^2 = (E[X])^2 * \lambda t$$

Example:

Customers arrive at a grocery store in a Poisson manner at an average rate of 10 customers per hour. The amount of money that each customer spends is uniformly distributed between \$8.00 and \$20.00. What is the average total amount of money that customers who arrive over a 2 h interval spend in the store? What is the variance of this total amount?

$$E[Y] = \frac{8+20}{2} = 14,\ \sigma_Y^2 = \frac{(20-8)^2}{12} = 12,$$

$$E[(X(t)]|_{t=2} = 2\lambda E[Y] = 2 * 10 * 14 = 280$$

$$\mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) = E(N)\mathrm{Var}(X) + [E(X)]^2\,\mathrm{Var}(N)$$

$$\sigma_{\mathrm{X(t)}}^2|_{t=2} = 2\lambda\sigma_Y^2 + (E[Y])^2\sigma_{X(t)}^2 = 2 * 10 * 12 + (14)^2 * 2 * 10 = 4160$$

Combinations of Independent Poisson Process

Also, known as merging of Poisson processes or superposition of Poisson Processes

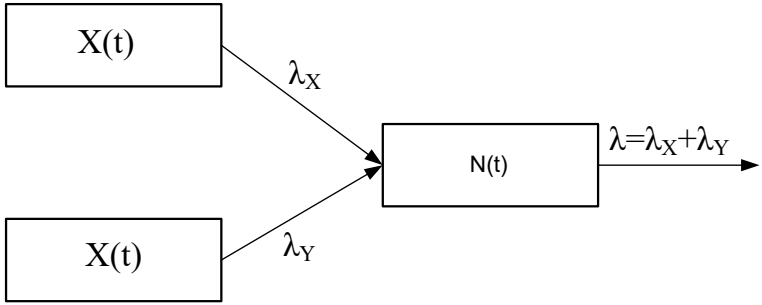
$\{X(t), t \geq 0\}$ and $\{Y(t), t \geq 0\}$ with rates λ_X and λ_Y are independent Poisson processes

Consider another process, $\{N(t), t \geq 0\}$

$$N(t) = X(t) + Y(t)$$

$$\begin{aligned}
P[N(t+\Delta t) - N(t) = 0] &= P[X(t+\Delta t) - X(t) = 0]P[Y(t+\Delta t) - Y(t) = 0] \\
&= [1 - \lambda_X \Delta t + o(\Delta t)][1 - \lambda_Y \Delta t + o(\Delta t)] \\
&= 1 - \lambda_X \Delta t - \lambda_Y \Delta t + o(\Delta t) = 1 - (\lambda_X + \lambda_Y)\Delta t + o(\Delta t) \\
&= 1 - \lambda \Delta t + o(\Delta t)
\end{aligned}$$

$\{N(t)\}$ is a Poisson process with rate $\lambda = \lambda_X + \lambda_Y$


X(t)
λX
X(t)
λY
N(t)
λ=λX+λY

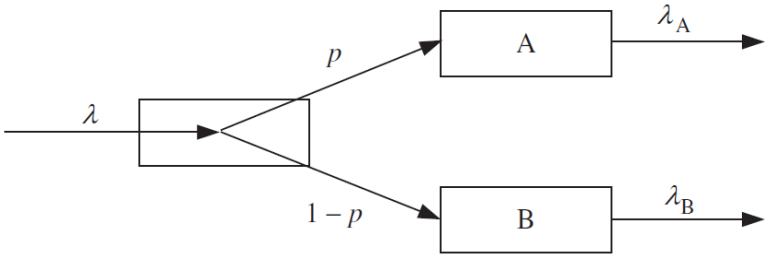
L_X, L_Y denote time until the first arrival.

$L > t$, if both $L_X > t$ and $L_y > t$. Also, L_X and L_Y are independent

$$P[L > t] = P[L_X > t]P[L_Y > t]$$
$$= e^{-\lambda_X t} e^{-\lambda_Y t}$$
$$= e^{-(\lambda_X + \lambda_Y)t} = e^{-\lambda t}$$

Where $\lambda = \lambda_X + \lambda_Y$ is the rate of the merged Poisson process

Subdivision of a Poisson Process



$\{X(t), t \geq 0\}$ is a Poisson process with rate λ

Divided into two processes A and B, arrival is of type A with prob. p and type B with prob. $(1-p)$

The arrival rate of A is $\lambda_A = \lambda p$

The arrival rate of B is $\lambda_b = \lambda(1-p)$

Both are Poisson processes

Comparison of First arrival Times of two independent Poisson Process

$\{X(t), t \geq 0\}$ and $\{Y(t), t \geq 0\}$ with rates $\lambda_X$and λ_Y are independent Poisson processes

Goal: To find the probability that the first arrival of X(t) occurs before that of $Y(t)$, i.e., $P[T_X < T_Y]$

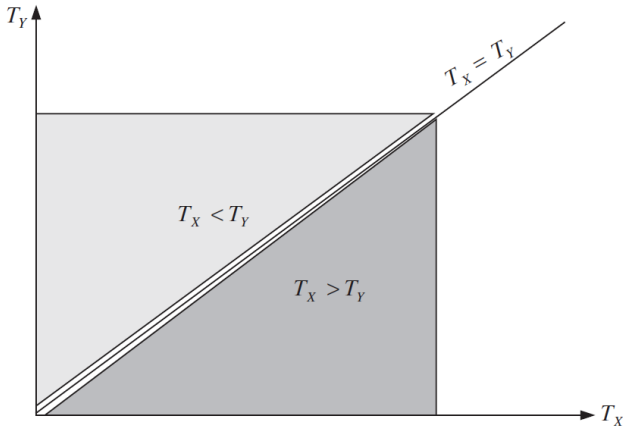
Note that T_X and T_Y are independent Exponential RVs

$$f_{T_X}(x) = \lambda_X e^{-\lambda_X x}, x \geq 0 \text{ and}$$

$$f_{T_Y}(y) = \lambda_Y e^{-\lambda_Y y}, y \geq 0$$

And, therefore

$$f_{T_X T_Y}(x, y) = \lambda_X \lambda_Y e^{-\lambda_X x} e^{\lambda_Y y}, x \geq 0, j \geq 0$$


T_Y
$T_X = T_Y$
$T_X < T_Y$
$T_X > T_Y$
T_X

$$P[T_X < T_Y] = \int_{x=0}^{\infty} \int_{y=x}^{\infty} f_{T_X T_Y}(x, y) \mathrm{d}y \, \mathrm{d}x = \int_{x=0}^{\infty} \int_{y=x}^{\infty} \lambda_X \lambda_Y \, \mathrm{e}^{-\lambda_X x} \, \mathrm{e}^{-\lambda_Y y} \mathrm{d}y \, \mathrm{d}x$$

$$= \int_{x=0}^{\infty} \lambda_X \, \mathrm{e}^{-\lambda_X x} \, \mathrm{e}^{-\lambda_Y x} \, \mathrm{d}x = \int_{x=0}^{\infty} \lambda_X \, \mathrm{e}^{-(\lambda_X - \lambda_Y) x} \, \mathrm{d}x$$

$$= \frac{\lambda_X}{\lambda_X + \lambda_Y}$$

Memoryless Property

$$P[X > s + t | X > t] = P[X > s], \qquad \text{for all } s, t \geq 0.$$

$$\frac{P[X > s + t, X > t]}{P[X > t]} = P[X > s]$$

Exponentially distributed RVs are memoryless

$$
\begin{aligned}
P[X > s+t] &= 1 - F_X(s+t) \\
&= 1 - (1 - e^{-\lambda(s+t)}) \\
&= e^{-\lambda(s+t)} \\
&= e^{-\lambda s} e^{-\lambda t} \\
&= P[X > s] P[X > t]
\end{aligned}
$$

$$
\frac{P[X > s+t]}{P[X > s]} = P[X > s]
$$

Fresh-start Property

The future states of the process after any time are independent of the past

Behave like a new Poisson process

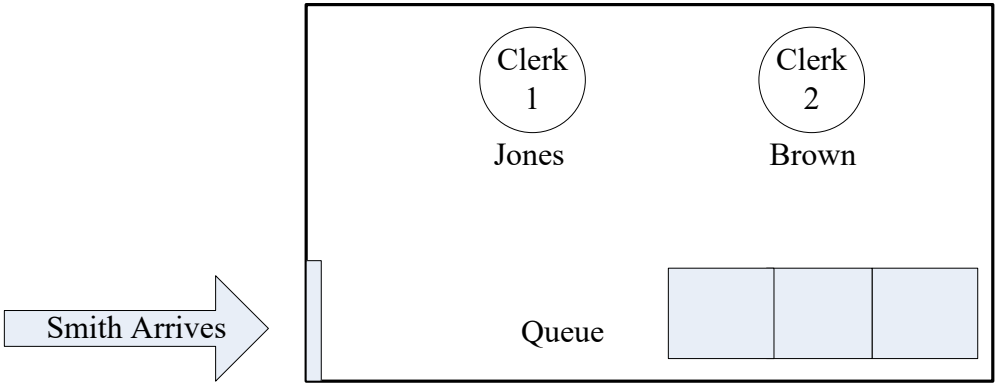
At any point of time, the Poisson process has the fresh start property

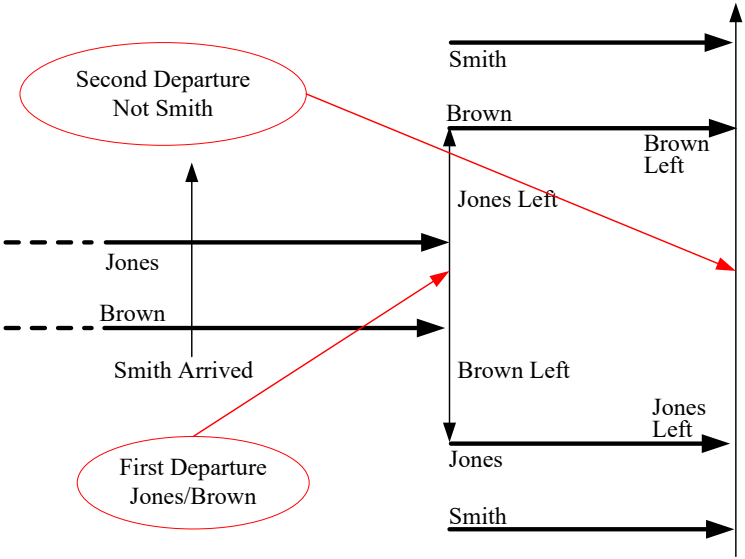
Example 2.3. Suppose that the amount of time one spends in a bank is exponentially distributed with mean ten minutes, that is, $\lambda = \frac{1}{10}$. What is the probability that a customer will spend more than fifteen minutes in the bank? What is the probability that a customer will spend more than fifteen minutes in the bank given that she is still in the bank after ten minutes?

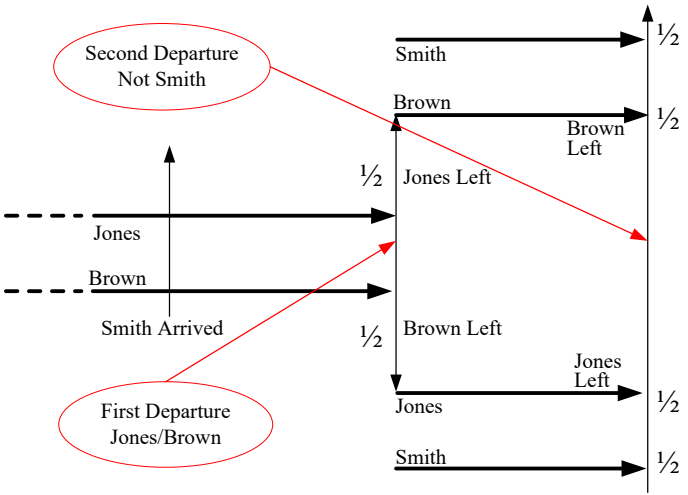
$$P[X > 15] = e^{-15\lambda} = e^{-3/2} \approx 0.220$$

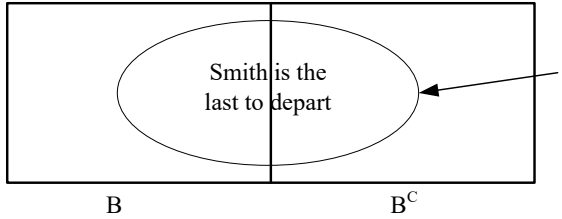
$$P[X > 5] = e^{-5\lambda} = e^{1/2} \approx 0.604$$

Example 2.4. Consider a post office that is run by two clerks. Suppose that when Mr. Smith enters the system he discovers that Mr. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Jones or Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with mean $1/\lambda$, what is the probability that, of the three customers, Mr. Smith is the last to leave the post office?




Second Departure
Not Smith
Smith
Brown
Brown
Left
Jones Left
Jones
Brown
Smith Arrived
Brown Left
Jones
Left
First Departure
Jones/Brown
Jones
Smith





B: Event that Jones left first

B^C: Event that Brown left first

A: Event that Smith is the last to leave

$$P[A] = P[A|B]P[B] + P[A|B^C]P[B^C]$$
$$= \tfrac{1}{2} * \tfrac{1}{2} + \tfrac{1}{2} * \tfrac{1}{2} = \tfrac{1}{2}$$

Example 2.4 [Contd.]. Now Suppose that the amount of time that a clerk spends with a customer is exponentially distributed with rate $\lambda_i, i = 1, 2$. Find the probability that Smith is the last to leave the post office.

Example 2.6. Let X_1 and X_2 be independent and exponentially distributed, with parameters λ_1 and λ_2, respectively. Find the expected value of $\min\{X_1, X_2\}$ and $\max\{X_1, X_2\}$.

$$E[T] = \frac{1}{\lambda_1 + \lambda_2}.$$

$$\begin{aligned} E[\max\{X_1, X_2\}] &= \frac{1}{\lambda_1 + \lambda_2} + \frac{\lambda_1}{\lambda_1 + \lambda_2}\frac{1}{\lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2}\frac{1}{\lambda_1} \\ &= \frac{1}{\lambda_1 + \lambda_2}\left(1 + \frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}\right). \end{aligned}$$

Example 2.5. Supose you arrive at a post office having two clerks at a moment when both are busy but there is no one else waiting in line. You will enter service when either clerk becomes free. If servcie times for clerk i are exponential with rate λ_i, $i = 1, 2$, find $E[T]$, where T is the amount of time that you spend in the post office.

$$Total\ Time = Waiting\ Time + Service\ Time$$

$$T = W + S$$

Let R_i, $i = 1, 2$ is the remaining service time of the i-th Clerk

Then, your waiting time is the minimum R_1 and R_2, i.e., when either of the two clerk is free, you start getting your service

$$
\begin{aligned}
E[T] &= E[\min(R_1, R_2) + S] \\
&= E[\min(R_1, R_2)] + E[S] \\
&= \frac{1}{\lambda_1 + \lambda_2} + E[S]
\end{aligned}
$$

$$
\begin{aligned}
E[S] &= E[S|R_1 < R_2]\frac{\lambda_1}{\lambda_1 + \lambda_2} + E[S|R_2 \leq R_1]\frac{\lambda_2}{\lambda_1 + \lambda_2} \\
&= \frac{2}{\lambda_1 + \lambda_2}
\end{aligned}
$$

Comparison of n-th and m-th arrival times of 2 independent Poisson processes

$\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$

$P(S_n^1 < S_m^2)$.

$$P(S_1^1 < S_1^2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$P(S_2^1 < S_1^2) = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^2$$

$$P\{S_n^1 < S_m^2\} = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n+m-1-k}$$

Bernoulli Process

Consists of a sequence of Bernoulli trials

The i-th trial produces a success with probability p,

$$P[X_k = 1] = p$$

and a failure with a probability

$$(1 - p), P[X_k = 0] = 1 - p$$

Example: sequence of coin tosses

Definition

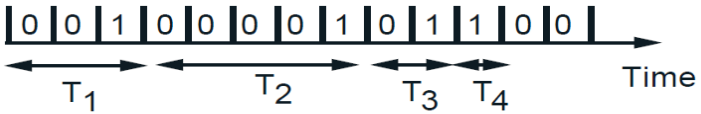
A discrete time CP $\{X_k, k = 0, 1, \dots\}$ is a Bernoulli process with probability of arrival in a time period is $p, 0 < p < 1$, if

1. $X_0 = 0$, when $k \leq 0$
2. X_k is integer valued and non-decreasing
3. Number of arrival in n time periods is binomially distributed

$$P[N_{m+n} - N_m = k] = \binom{n}{k} p^k (1-p)^{n-k}; \qquad k = 0, 1, \dots, n.$$

4. X_k has stationary increment
5. X_k has independent increment

Interarrival times



T_1 is the first arrival time

T_k is the interarrival time of the k-th event

Time gap between the arrival time of the k-th and $(k-1)$-th events

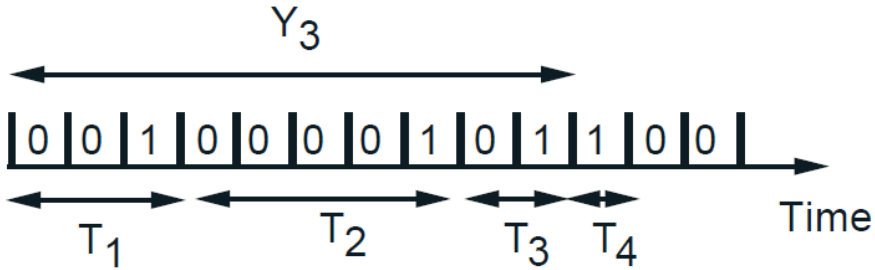
T_k, $k = 1, 2, \ldots$ are the interarrival times of the successive events

T_k is distributed as

The PMF of T_k is

$$P[T = t] = p(1-p)^{t-1}; \qquad t = 1, 2, \ldots$$

k-th Arrival Time



$$T_1 = Y_1, \qquad T_k = Y_k - Y_{k-1}, \qquad k = 2, 3, \ldots.$$

Memory-less Property

The memoryless property of a RV variable is defined as

$$P[X > s + t | X > t] = P[X > s], \qquad \text{for all } s, t \geq 0.$$

$$P(X > s + t) = P(X > s)P(X > t)$$

For the geometric RV

$$
\begin{aligned}
P[X > s+t] &= 1 - F_X(s+t) \\
&= 1 - (1 - (1-p)^{s+t}) \\
&= (1-p)^{s+t} \\
&= (1-p)^s (1-p)^t \\
&= P[X > s] P[X > t]
\end{aligned}
$$

$$P[X > s+t | X > t] = P[X > t].$$

<u>Fresh-start Property</u>

Starting from any time period

X_k is a new CP, due to

1. the memory-less property X_k forgets everything before k
2. independent increment property
    a. # of #arrivals starting from k, independent of # of arrivals before k

$X_k, k = l, l+1, l+2, \ldots$ is a new CP as compared to

$X_k, k = 1, 2, \ldots, l-1$

r-th Arrival time T_r

Y_r is the time periods to have the r-th arrivals

$Y_r = T_1 + T_2 + \cdots + T_r$

Sum of IID geometric random variables

The mean and variance of Y_k are given by

$$E[Y_k] = E[T_1] + \cdots + E[T_k] = \frac{k}{p},$$

$$Var[Y_k] = Var[T_1] + \cdots + Var[T_k] = \frac{k(1-p)}{p^2}$$

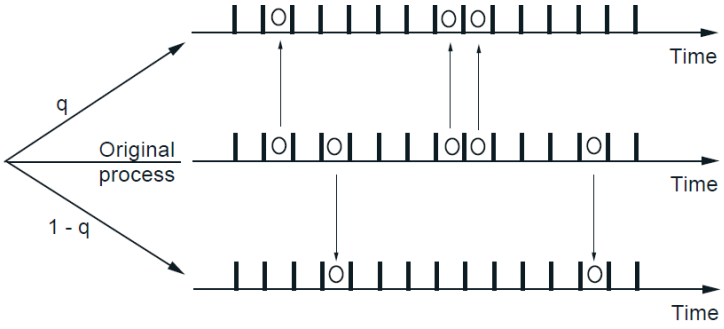
Y_r is distributed as

$$Y_r \sim pascal(r, p)$$

PMF of Y_r is

$$P_Y(k) = \begin{cases} \binom{t-1}{k-1} p^k (1-p)^{t-k}, & \text{for } t = k, k+1, k+2, \ldots; \\ 0, & otherwise. \end{cases}$$

Splitting and Merging of BPs



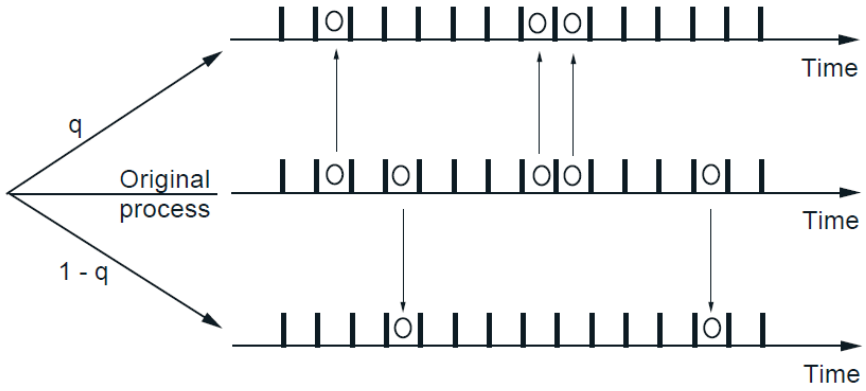
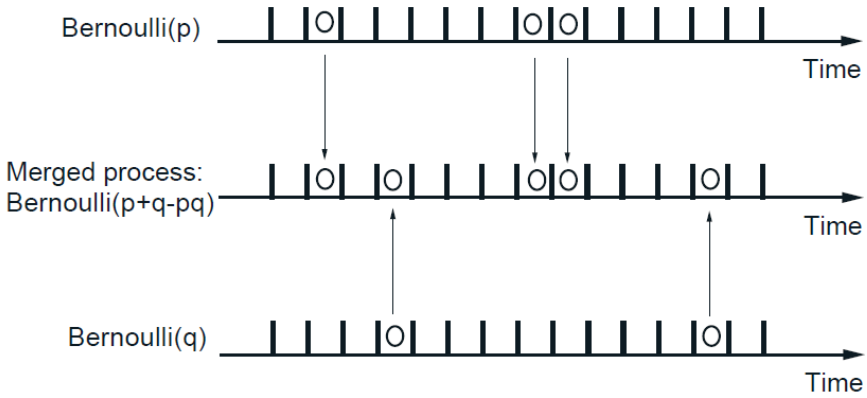



Figure 2.3: Splitting of a Bernoulli process.